What the Policy Sees

Observation of the CALM speed recommender

1 When the policy observes

The policy decides once per bump. The observation it receives is taken at the moment the slowdown window opens, 25 m before the bump, while the car is still cruising at about the scenario speed. Everything the policy needs to choose a depth must therefore be in this one vector: how the car is moving, and what the road ahead looks like.

2 Layout

The observation has 23 numbers: 8 describe the vehicle and 15 describe the road ahead (Table 1).

3 Road ahead

The preview looks $D = 40\,\mathrm{m}$ ahead (Figure 1) and lists the bumps whose start lies within this distance and whose end the car has not yet passed, nearest first, at most three. Holes are not listed. For a bump with start distance d, width L and height A the four values are scaled to $[0, 1]$:

$$\frac{d}{40\,\mathrm{m}}, \qquad \frac{L}{8\,\mathrm{m}}, \qquad \frac{A}{0.15\,\mathrm{m}}, \qquad \frac{v_{\mathrm{top}}/L}{10\ \mathrm{Hz}}, \tag{1}$$

each clipped to $[0, 1]$. The last value is the frequency at which the wheel meets the bump at the top speed v_{top}, the larger of the scenario speed and the speed floor; it is the quantity that decides whether the bump excites the body or the wheel. A bump that is not present is filled with $(1, 0, 0, 0)$: far away, no width, no height, no excitation. The speed ratio is v/v_{top} and the distance since the last bump is set to 1 when no bump has been passed.

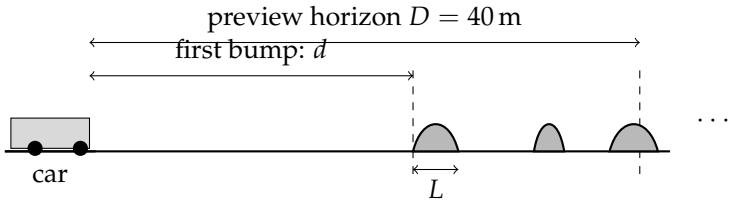


Figure 1: Geometry of the preview. Bumps beyond D are not visible.

4 Perception

The baseline observes the true bump geometry of the road: no noise is added and no camera detector is in the loop. The policy is therefore trained and scored as if the road ahead were detected perfectly. A camera-based detector with measured range bias and noise exists as a separate option

No.	Quantity	Definition
Vehicle (8)		
1	Front wheel travel	front wheel height minus body height (m)
2	Wheel acceleration	vertical acceleration of the wheel (m/s^2)
3	Body acceleration	vertical acceleration of the body (m/s^2)
4	Pitch acceleration	angular acceleration of the pitch (rad/s^2)
5	Pitch angle	body pitch (rad)
6	Speed	current speed v (m/s)
7	Longitudinal acceleration	acceleration of the last step (m/s^2)
8	Speed error	reference speed minus current speed (m/s)
Road ahead (15)		
9–12	Nearest bump ahead	distance, width, height, frequency (see below)
13–16	Second bump ahead	same four values
17–20	Third bump ahead	same four values
21	Speed ratio	v/v_{top}
22	Distance since last bump	distance to the end of the last bump passed, divided by 40 m
23	On-bump flag	1 while a bump is under the car, else 0

Table 1: The 23 observed values.

of the environment (replay of recorded detections, or a noise model fitted to them); it is not used in the baseline.

5 Normalisation

During training the observation is normalised with a running mean and standard deviation per value and clipped to ± 10. The statistics are saved with the policy in `vecnormalize.pkl`; the policy cannot be used without them.

6 Limits

The policy sees at most three bumps within 40 m, so a bump farther away does not influence the decision. Because the observation is taken when the window opens, the vehicle values carry little

information; most of the decision rests on the preview of the next bump and the scenario speed.